

The distributive law



1 A student incorrectly used the distributive property and wrote $6(7x + 2) = 42x + 2$. Explain how to correct the error.

2 Expand the following expressions:

- | | | | |
|------------------------|--------------------------|----------------------------|-----------------------------|
| a $5(s + 7)$ | b $9(5 + w)$ | c $3(r - 6)$ | d $(y + 8) \times 7$ |
| e $-8(t + 10)$ | f $-8(c - 5)$ | g $-(y + 2)$ | h $4(4s + 5)$ |
| i $-10(8 + 3r)$ | j $-8(-2t - 3)$ | k $p(q + 4)$ | l $9(u - v)$ |
| m $-(2m - 3n)$ | n $-r(11s + 13t)$ | o $4(2r + 3s + 5t)$ | p $6(4m + 3n + 5p)$ |

3 Expand the following expressions:

- | | | | |
|---------------------------|---------------------------|----------------------------|----------------------------|
| a $6m(2n + 3)$ | b $9(v^2 + 6)$ | c $r(r + 7)$ | d $6m(4m + 7)$ |
| e $-6a(2a + 9)$ | f $4wy(y + w)$ | g $4p(4q - 3 + 6r)$ | h $2f(f - 7g - 6h)$ |
| i $-10(y^2 + 7)$ | j $-(9 - s^3)$ | k $r(s^2 + 8)$ | l $-10(3 + 8u^3)$ |
| m $4s^5(4t^4 - 9)$ | n $6p^3(2q^5 - 9)$ | o $9f(f - 4g - 6h)$ | p $3uv(v + u)$ |

4 Expand and simplify the following expressions:

- | | |
|------------------------------|----------------------------|
| a $4(x + 8) - 2$ | b $7 + 5(3x + 4)$ |
| c $5 + 3(x + 4)$ | d $8(3x + 4) - 6x$ |
| e $34 + 9(4x - 5)$ | f $5x - 8(2x - 3)$ |
| g $9x - 5(2x + 3)$ | h $-8(5x - 7) - 9$ |
| i $4y + 5 + 6(y - 9)$ | j $4(5(x - 3) + 2)$ |

5 Expand and simplify the following expressions:

- | | |
|--------------------------------------|--------------------------------------|
| a $x(x + 4) + 8(x + 9)$ | b $8y^2 - 4y + 7y(y - 5)$ |
| c $3(6x - 5y) - 2(9y + 4x)$ | d $7(y + 9) + 5(y + 6)$ |
| e $10(x - 4) + 3(x + 8)$ | f $5(y + 7) - (y - 9)$ |
| g $-8(x + 4) - 3(x - 2)$ | h $11(a + 4) - 6(4 - a)$ |
| i $4u(6u + 9v) - 2u(8v + 5u)$ | j $9w(8w + 6z) - 5w(4z + 2w)$ |

Applications

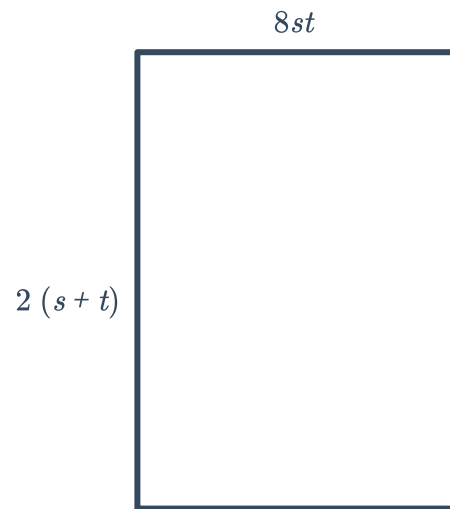
6 A solar energy company offers a range of solar panels, each with a power rating of

$(m + 8)$ kW, where m is the panel model number. Write an expression in terms of m for the total amount of power produced by 11 solar panels.



- 7 The representatives of 8 countries are attending a global summit, and each country is able to send $(2r - 3)$ representatives. Write an expression for the total number of representatives attending the summit.
- 8 A forklift operator has enough time to make 5 trips between the two ends of a warehouse. A forklift can carry $(2 + 3n^4)$ boxes at a time. Write an expression for the total number of boxes that can be carried across the warehouse.

- 9 Consider the rectangle shown:
Find an expression for the area of the rectangle in expanded form.



- 10 A rectangle is 8 m longer than it is width. Its width is represented by the pronumeral u . Form an expression for the area of the rectangle in expanded form.

Factorisation

- 11 State whether the following are factors of $33wy$:
- | | | | |
|----------|----------|--------|--------|
| a 11 | b $66wy$ | c yw | d $3w$ |
| e yw^2 | f 33 | | |
- 12 Find the highest common factor of the following groups of numbers:
- | | |
|----------------|---------------------------|
| a 2 and $6x$ | b 3 and $15x$ |
| c $6x$ and 21 | d 10 and $16u$ |
| e $35x$ and 14 | f $4x$ and $16xy$ |
| g 15 and $5x$ | h $24m$, $12n$ and $18q$ |



 **13** Complete the factorisation of the following expressions:

a $2a + 16 = \text{ⓐ}(a + 8)$

b $5y - 30 = \text{ⓑ}(y - 6)$

c $24n + 15 = \text{ⓒ}(8n + 5)$

d $16a - 6 = \text{ⓓ}(8a - 3)$

e $6v + 18 = 6(\text{ⓔ})$

f $9a - 36 = 9(\text{ⓕ})$

g $24a + 9 = 3(\text{ⓖ})$

h $6w - 27 = 3(\text{ⓗ})$

i $4r + 12 = \text{ⓔ}(r + 3)$

j $5v - 30 = \text{ⓙ}(v - 6)$

k $18c - 24 = \text{ⓚ}(3c - 4)$

l $2r + 14 = 2(\text{Ⓛ})$

m $5t - 10 = 5(\text{Ⓜ})$

n $54u + 12 = 6(\text{Ⓝ})$

14 Factorise the following expressions:

a $6v + 30$

b $6a - 48$

c $45t - 40$

d $2m + 10n$

e $54 - 60fg$

f $8g - 72h$

g $9f + 15g$

h $3w + 15$

i $2c - 18$

j $40c - 35$

k $6w + 30y$

l $14y + 4w$

m $4a - 14b$

n $32 - 36jk$

o $8jkt - 18$

p $3fg - 10fh$

q $3rtw - 8rw$

r $4rt - 6r^2t$

s $16wy - 36w^2y$

 **15** Complete the following factorisation:

$$2w - 10z + 8y = 2(\text{ⓐ} - \text{ⓑ} + \text{ⓒ})$$

16 Factorise the following completely:

a $8g + 28h - 20$

b $4w + 4z + 4y$

c $4u - 12v + 20w$

d $10m - 25n - 15p$

e $3m - 6n + 12p$

f $10g - 35h + 15$

g $zw - zy - zk$

h $60r - 25rt - 35r^2t$

i $12m - 32mn - 20m^2n$

j $16z - 32yz - 24y^2z$

k $25r^2t - 15rt^2 - 45r^2t^2$

Factorisation of negatives

17 Factor out the negative coefficient of the following expressions:

a $-4r - 3s$

b $-7r - 7s$

c $-2 - 9r$

d $-8x - 5y$

18 Factorise the following expressions:



a $-8b - 16$

b $-18a + 16$

c $-80v - 30$

d $-36 - 6x$

e $-25u - 35v$

f $-35m + 49n$

g $-8pqr - 6pr$

h $-8w^2 + 3w^2y$

i $-24j^2 - 27j^2k$

j $-8m^2 + 9mn$

k $-10u^2v + 9uv^2$

l $-21j^2 - 36j^2k$

m $-2s - 10$

n $-8v^2 + 56$

o $-12s + 10$

p $-28b - 21$

q $-63r^2 + 28$

r $-4m - 32n$

s $-49y - 21w$

19 Factorise the following completely:

a $-18p - 9q - 45r$

b $-10r - 25s + 20t$

c $-20u - 25v + 45w$

d $-8w - 18y - 10z$